JOEY HUANG

Research Interest

2024-2026
- **Graph Machine Learning**
- Information Retrieval

2026-2029 expected
- **Quantum Information**
- Quantum Logic Synthesis and Optimization
- GNN-RL for Quantum Circuit Compilation

Education

2026-2029 **Bachelor of Science**, *Department of Computer Science*, National Tsing Hua University, Taiwan

2024-2026 National Chengchi University, Taiwan
Major: Undeclared Major in College of Communication
Double Major: Bachelor Program in Artificial Intelligence Applications
Minor: Department of Mathematical Sciences

2021-2024 National Hsinchu Senior High School, Taiwan

Work Experience

2026/7-2026/8 **R&D Intern** (Information Retrieval), Taiwan Dubao

Research Experience

2026/2 **Integrating Geometric Graph Embeddings with Large Language Models for Reasoning in Occupational Safety Regulations**
Hsu Kuan, Huang. National Science and Technology Council Undergraduate Research Program, Division of Intelligent Computing, Department of Engineering (115-2813-C-004-067-E); Presented at 2025 GDG NCCU Final Presentation Meeting.

2026/9 **Fraud Detection Based on Dual-Channeled Signed Graph, Laplacian Regularization and Gradient Boosting**
Hsu Kuan Huang , En Shuo Zhang , Chao-Ling Chen *. Written for TANET. Presented at 2026 GDG NCCU Final Presentation Meeting.

Invited Lectures

2026/11 **Research Talk on *LLM4Code: Understand How AI Learns to Code***
Google Developer Group (GDG) at National Chengchi University

2026/3 **Research Talk on *Graph Neural Networks and Knowledge Graphs: From Basic Theory to Geometric Constraint Generation***
Google Developer Group (GDG) at National Chengchi University

Relevant Coursework

Mathematics
- Linear Algebra (87)(93)
- Calculus A (90)(99)
- Statistics (I) (90)

Computer Science
- **Algorithms (93)**
- Data Structure (88)
- Stochastic Processes for Networking
- Social Networks

Machine Learning
- **Graph Neural Network** (94.25, audit)
- Natural Language Processing (99)
- Special Topics in Artificial Intelligence Applications (95)
- Capstone: Practice of Generative AI and Data Science (97.6)
- Machine Learning: Concepts and Applications (91)
- Artificial Intelligence Programming (100)

Quantum Information and Physics
- **Introduction to Quantum Computing**
- Circuits and Electronics (I)
- **Introduction to Nano Science and Technology (96)**
- General Physics (100)(87)

Awards and Competitions

2025/4 Finalist, AIWave: Taiwan Generative AI Applications Hackathon

2026/8 Semi-final Round, DevJam (GDG on Campus Hackathon)

2026/11 Finalist, AI for Cross-Disciplinary Sustainability Innovation Competition
AI Center, National Chengchi University

2027/1 **Recipient, NCCU AI Interdisciplinary Research Proposal Grant**
AI Center, National Chengchi University

Leadership and Service

2022/9-2023/8 **Club President**, 9th Session of Hsinchu High School Model United Nations

2022/10 CCMUN Conference (Awarded Best Position Paper)

2023/1 Chief Organizer and Chair, Hsinchu District Five-School MUN Winter Training

2023/4 **Secretary General**, Fong MUN Conference

2023/4 **Head Organizer and Chair**, Ten-School MUN Spring Training

2023/6 Academic Advisor and Chair, RsMUN Conference

2023/9-2024/8 Faculty Advisor, 10th Session of Hsinchu High School Model United Nations

Professional Development

2025/3 CS50's Introduction to Artificial Intelligence with Python

2025/8 Python for Quantum Computing (Packt, Coursera)

Technical Skills

Machine Learning
- Pytorch
- Transformers
- Scikit-learn
- Pytorch Geometric

Quantum Computing
- Qiskit

Language
- Mandarin (Native),
- English (TOEIC 905),
- Japanese (**JLPT N1**, PR 96.8)

Programming
- Python
- C++